1. Frobenius as a symmetry of the decomposition search: audit and prepared m-sweep

Campaign directory `research/frobenius_msweep_20261027/`. 2026-10-26 to 27. This is a search report: an audit of a closed lead and the preparation of five measurement cells. **No experiment has been run and no data are presented.**

1.1. Question

LEAD-ICB-frobenius-search-symmetry, in the crypto-autoresearcher atlas under `knowledge/leads/ecdlp/ic-binary/`, is closed. Its stated obstruction has five parts, C1–C5. The owner's challenge was that the area was examined only at m = 2 or 3, where m is the number of factor-base points per decomposition, while chained encodings make m up to about 20 reachable. The question is whether that closure actually covers m ≥ 3.

1.2. Method

An audit workflow (`wf_3231c71e-f6b`, 13 agents, read-only) ran in three stages:

- one auditor per sub-claim re-read the cited ledger evidence and code;
- an m-coverage census listed 25 prior experiments;
- a gap synthesis was followed by three idea generators, each with an adversarial red-team.

The full JSON is kept with this campaign as `audit_20261026.json`. The coverage figure is drawn from the recorded per-claim values in `coverage.json`. Status of each statement: *derived* (audit argument), *recorded* (prior evidence), *prepared* (design only).

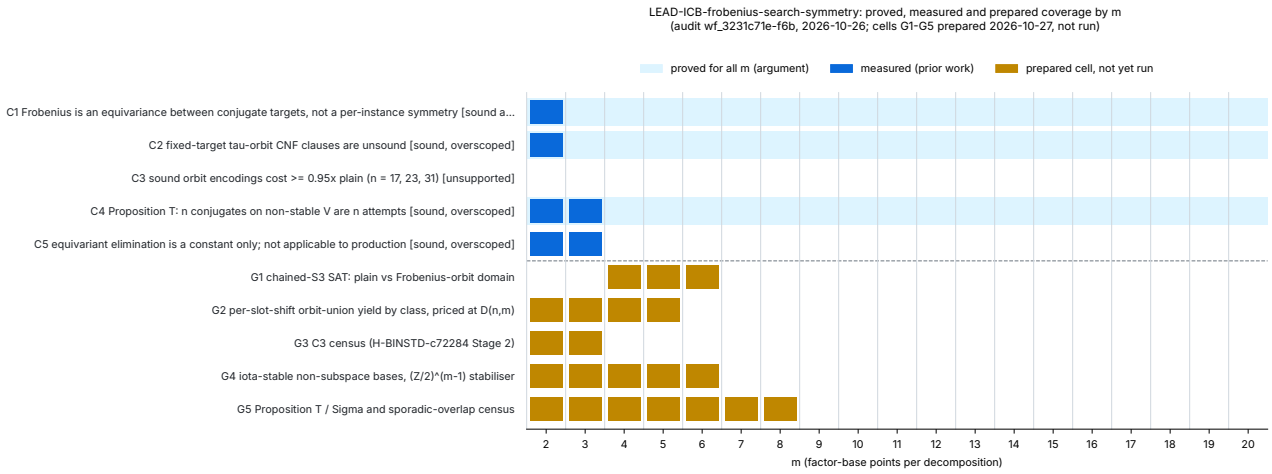


Figure 1: Proved, measured and prepared coverage of C1–C5 by m. Sources: the audit (`wf_3231c71e-f6b`) and the cell protocols. Prepared cells (gold) have not been run.

1.3. Findings of the audit

- **C1, C2, C4 are proved for all m** (*derived*).
  - Frobenius maps the ideal of target R to the ideal of σR, because S_{m+1} has F_2 coefficients. This holds uniformly, including for chained encodings with intermediate points.
  - Independent per-slot shifts are never a symmetry of a fixed-target instance. That would need $\Sigma(\lambda^k_i - 1)P_i = 0$.
  - Experiments behind C1 and C2 ran at m = 2 only (n = 17, 19). They are close to tautological checks of the lemma.
- **C3 is unsupported** (*recorded*). The ≥ 0.95× figure is a pre-registered threshold in H-BINSTD-c72284, and its Stage 2 solver census was never authorised or run.
- **C5 is measured only on the m = 2 barrel** (*recorded*), at 2.15× / 3.15× over gf2_elim for n = 11 / 13. The production pipeline builds no σ-symmetric system, so "not applicable" is structural, not a measured regression at m ≥ 3.
- **No Frobenius-symmetry solver experiment exists at m ≥ 3** (*recorded*, census of 25 cells). The largest m with any Frobenius structure is the 6-sum extraction (m = 4–6, n = 23–41) and the m = 10 rotated chain at n = 83, which has no committed solve.
- **The m = 2 orbit-union cost comparison was mispriced** (*derived*). It was priced against n^{m-1} systems, but the exact count of distinct systems is D(n,m) = C(n+m−1, m)/n, about $n^{m-1}/m!$. The gap is a factor of order m! and grows with m.
- **A genuine m-growing stabiliser exists** (*derived*). Translation by the 2-torsion point, ι: x ↦ √b/x, is a same-instance stabiliser of order 2^{m-1}, but only on ι-stable bases. Linear subspaces cannot be ι-stable at n = 131, so only non-subspace bases remain.
- **Red-team** (*derived*): 12 new ideas were generated and none survived as a lever. Most were relabellings, amortisation, or failed the decisive count.

1.4. Prepared cells (not run)

Each cell has a frozen PROTOCOL.md with dated amendments, native Rust code, unit tests, a RUN_CARD.md and a README. Each went through two independent review rounds and three repair passes. All 148 unit tests pass.

cell	question	m	tests
G1	chained-S3 SAT: plain vs exact Frobenius-orbit domain; C3 grade	4–6	25
G2	orbit-union yield by shift-pattern class, priced at D(n,m)	2–5	30
G3	the never-run C3 census (H-BINSTD-c72284 Stage 2)	2–3	38
G4	ι-stable non-subspace bases, $(Z/2)^{m-1}$ stabiliser	2–6	24
G5	Proposition T / Σ and sporadic-overlap census	2–8	31

1.5. Why nothing was run

The repository requires a resume checkpoint in S3 before any experiment. This container has no upload credential, so the runs wait for a session that has `CHECKPOINT_UPLOAD_TOKEN`, or for the Iso Lab EKS route.

The repository's `main` also did not compile: five broken merge resolutions. They are repaired in PR #2675 and carried on this branch.

1.6. Scope and limits

- This report presents no measured result and no change to cost or the leaderboard.
- The prepared cells stop at m = 8 (see the figure). Coverage for m = 9–20 still needs a chained encoding at those sizes. The m = 10 rotated chain at n = 83 is the only existing code there.
- The cells' authors expect nulls in most cells. A measured null at $m \geq 3$ that replaces an asserted closure is itself the deliverable.